

Men may be
from Mars, and
Women from Venus;
but everyone
is certain to
converge on
this planet!



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Honey, I Nuked the MBR

Playing with the MBR is dangerous. If you mess up, you could render your system incapable of booting into any operating system. So how do you prevent this from happening? Take a look at some preventive measures and cures to this problem.

First, plan ahead for such contingencies. Make backups and create boot diskettes for all the operating systems that you are running, and keep the original bootable install CDs handy as well. If you managed to wipe out the MBR, or render the NTLDR or LILO boot loader unable to boot, we can help you boot into Windows. To boot into Linux, you'll have to re-install LILO.

The tools you need before you start	What to do if you nuked your MBR by mistake
Windows 98 Create a Windows 98 bootable diskette or MS-DOS bootable diskette	Boot from the diskette and run the following command from the DOS command line: fdisk /mbr , then press [Enter] and reboot.
Windows XP The bootable Windows XP installation CD	Boot from the CD and run the recovery console, select the installation and get into command line interface, type 'fixmbr' to fix the MBR, confirm, then exit and reboot.
Linux Bootable Linux diskette or Linux installation CD	Boot into Linux using the diskette or the CD. Log in as root. At the command prompt, use /sbin/lilo -u to uninstall LILO. If you're using the CD, get into Rescue Mode (for Red Hat Linux, press [F5] at the boot prompt). Then go through the steps, and finally, to make your system the root partition, type 'chroot /mnt/sysimage' . This is essential to run LILO if you've booted from the CD.

```
[operating systems]
multi(0)disk(0)rdisk(0)partition(1)\WIN-
DOWS="Microsoft Windows XP Professional"
/fastdetect
C:\bootsect.lnx="Linux"
```

Now reboot the system. NTLDR will prompt you to choose between Windows XP—which is default—and Linux. If you choose Linux, LILO will kick into action and present you with its boot menu—this again gives you the option of booting into Linux or DOS. Selecting DOS will take you back to the NTLDR boot menu.

Professionals to boot

Beneath its rather simple interface, LILO is a very powerful boot loader. To go beyond the graphical boot menu and take a look at the text menu, press **[Ctrl] + [X]** at the graphical boot screen. A LILO prompt comes up. You can browse through the available options by pressing [Tab]; this will list out all the options. At the LILO prompt, you will have to key in the option, and any parameters that you would like to put in.

Should you wish to log into single-user mode as root after losing the root password, or to repair a damaged system, type **'LILO:linux single'**. This boots Linux into a single user state. To mount another partition, use the **LILO: root=/dev/hdaX** command, where 'X' is the partition you want to

mount—an 'ro' option can also be added, if you want the partition to be mounted as read-only. You can also get LILO to recognize additional memory if added to the system. This will generally be recognised, but should you have trouble, you can add **linux mem=256M** (if you have 256 MB of RAM).

Consult the LILO man pages and infopage for more esoteric options. If there are some options that you pass regularly, such as getting Linux to recognize additional memory, append **append="mem=256M"** to the bottom of the required image.

Finally, be sure to run LILO once afterwards, to reflect all changes. Remember that LILO creates a map file at installation time, and doesn't read the configuration file at boot-up. So, as superuser, run **/sbin/lilo** at the prompt. If all goes well, you will see an output without error messages or warnings. Most fatal errors can be traced back to entries in the **/etc/lilo.conf** file.

Though this article is LILO-specific, it also shows you how boot loaders work in general. Just remember, while LILO (and GRUB) are great fun to tweak, they can also cause you quite a lot of anxiety if you mess up. Keep your boot diskettes handy, be cautious, but explore and make sure you lace up your boots just right! ■

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